

# Impact of pharmaceutical care on adherence, hospitalisations and mortality in elderly patients

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**Abstract** *Background* Elderly polypharmacy patients may be more at risk of not adhering to medication. If so, the underlying reasons may be more readily disclosed during private discussions with patients. Hence pharmaceutical care discussions at home might improve treatment adherence. *Objective* The aim of this study was to investigate the impact of pharmaceutical care on medication adherence, hospitalisation and mortality in elderly patients prescribed polypharmacy. *Setting* Pharmaceutical care discussed at home. *Methods* A randomised controlled trial with two arms; pharmaceutical care ( $n = 315$ ) and controls ( $n = 315$ ) was designed. It involved patients aged 65+ years living in Aarhus, Denmark who used five drugs or more without assistance. Pharmacists visited the pharmaceutical-care patients at home, once only, and followed them during the subsequent year with three telephone calls. Non-adherence was measured by a pill-count. Patients were categorised as non-adherent if their mean adherence rate for all drugs consumed was  $<80\%$ . The impact of pharmaceutical care on non-adherence and hospitalisation was analysed by  $2 \times 2$  tables, and mortality by Cox regression. *Main outcome measure* Medication adherence, hospitalisation and mortality. *Results* The final analyses included 517 patients (median age 74 years; females 52 %). Dropouts were more frequent for the pharmaceutical-care group than for controls. Pharmacists encountered drug-related problems amongst 72 % of pharmaceutical-care patients. Pharmaceutical-care patients (11 %) and control

patients (10 %) were similarly nonadherent (Odds ratio 1.14; 95 % confidence interval 0.65–2.00), and similar with respect to hospitalisation frequency (30 vs. 28 %; Odds ratio 1.14; 95 % confidence interval 0.78–1.67) and mortality (7.5 vs. 5 %; Hazard ratio 1.41; 95 % confidence interval 0.71–2.82). *Conclusions* Pharmaceutical care given to our elderly polypharmacy patients made no significant impact on medication adherence, hospitalisation or mortality, when compared to comparable control patients.

**Keywords** Denmark · Drug-related problems · Elderly · Hospitalisation · Medication adherence · Mortality · Pharmaceutical care · Polypharmacy

## Impacts on practice

- Pharmaceutical care can be an strategy to identify drug-related problems in elderly patients
- Patients adherence to medicine, hospitalisation and mortality do not appear to be improved by complex interventions based on pharmaceutical care principles
- We advocate that pharmaceutical care must be better developed and investigated before being implemented

## Introduction

‘Pharmaceutical care’ is the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient’s quality of life [1]. Pharmaceutical care involves the process through which a pharmacist cooperates with a patient and other professionals in designing, implementing, and monitoring a therapeutic plan that will produce specific therapeutic outcomes for the patient. This in turn

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involves three major functions: (1) identifying potential and actual drug-related problems (DRPs), (2) resolving actual DRPs, and (3) preventing potential DRPs [1].

Pharmaceutical care applies at pharmacies, general practitioners' clinics, a patient's home, and hospitals. The need for such care at home arises because many elderly patients are homebound and certain DRPs are more readily detectable in a home setting [2].

The impact of pharmaceutical care has been evaluated in different populations using different endpoints. The impact of home-based pharmaceutical care on prescribed treatment adherence in the elderly varies. Five [3–7] of eight studies [3–10] found positive effects on adherence to long-term treatment. The interventions were all complex with different designs of home-based pharmaceutical care and different numbers of visits to the patients. It is not clear which part of the intervention that was the most successful.

Ageing itself does not predict non-adherence. However, associated risk factors do increase with age, e.g., multiple chronic medical conditions necessitating several medicines, multiple dosage schedules, and more frequent administrations [11–13].

Pharmaceutical care at home did not impact on hospitalisation rates in four [8, 14–16] of six studies, while two other studies reported contrary significant or borderline significant [4] effects, i.e. respectively a decrease in unplanned readmissions from 45 % of the patients in the control group to 28 % in the intervention group ( $p = 0.05$ ) [4] versus an increase with a total of 178 emergency readmissions in the control group and 234 in the intervention group ( $p = 0.009$ ) [17]. Four studies observed similar mortality rates [4, 8, 14, 17]. The cited studies recruited participants variously from hospitals and general practice, some focused on very old or high-risk patients [3–10, 14–17]. A review of pharmacy home services by MacKeigan and Nissen [2] concluded that the overall evidence of impact on treatment adherence is limited despite more rigorous recent methods of evaluation.

By contrast, the present randomised controlled trial sought evidence for a possible impact of pharmaceutical care on treatment adherence amongst 'healthy' elderly polypharmacy patients aged 65+ living at home. Such patients have not been studied before in this respect. The present study involved a home visit by a community pharmacist followed by three telephone calls during a single year. Telephone calls were chosen as two previous studies [18, 19] had shown positive effects of telephone counselling by pharmacists on treatment adherence.

## Aim of the study

The present aim was to investigate the impact of pharmaceutical care on medication adherence, hospitalisation and

the mortality of home-living elderly (65+) patients prescribed polypharmacy.

## Methods

Data were collected as part of a major randomised controlled trial 'Methods for Improving Compliance with Medicine Intake' (MICMI) [20]. The principal aim of the latter project was to examine whether 'pharmaceutical care' or an 'electronic reminder system' would improve treatment adherence in a population of elderly patients prescribed polypharmacy. The present analysis evaluated solely the impact of pharmaceutical care.

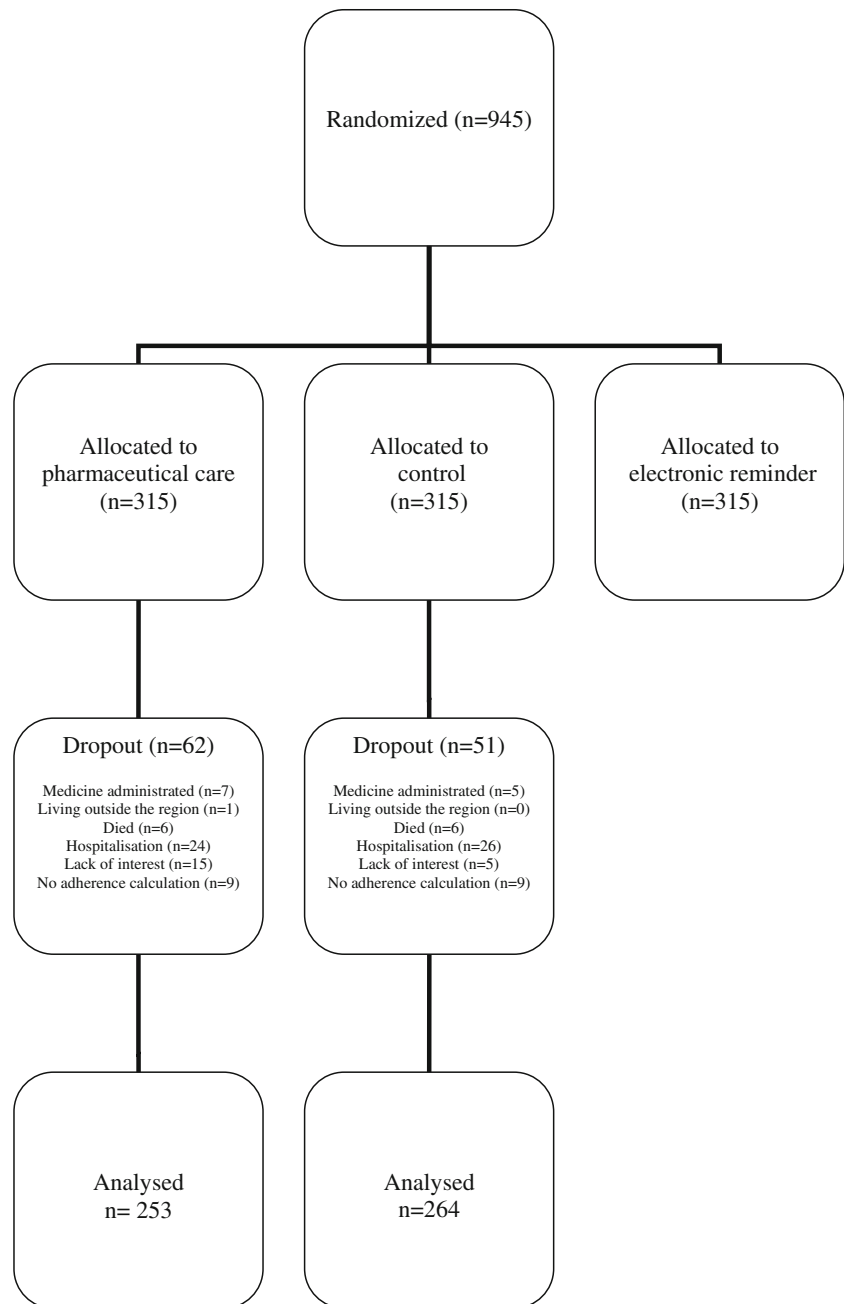
## Participants

All patients lived at home in the municipality of Aarhus, Denmark, and were identified in the National Health Insurance Population Register. Patient inclusion criteria were age  $\geq 65$  years and at least five current prescription drugs taken without assistance. The patients were contacted by letter and a follow-up telephone call. Patients with the following characteristics were excluded: residence in a nursing home, terminal illness, cognitive disorders such as dementia, medication supervised by healthcare providers, immigration to Denmark after January 2005, and severe motor impairment.

A total of 945 patients were included in the MICMI study (Fig. 1). Patients hospitalised for more than 7 days during the study were excluded before the final adherence evaluation. Details of the sampling procedure have been fully described elsewhere [21]. Patients were randomly assigned to three subgroups, i.e. two intervention groups (a 'pharmaceutical care' subgroup and an 'electronic reminder system' subgroup) and a control subgroup. Control patients were not subject to any intervention. The 'electronic reminder system' subgroup will not be considered here.

## Pharmaceutical care intervention

Patients in the 'pharmaceutical care' group were visited at home by a pharmacist at the beginning of the project. The pharmacist examined the medicines list with regard to possible side-effects, interactions, and administration, then tried to make the regime less complex, informed the patients meanwhile about the drugs, listened to questions concerning the drugs, handed over information leaflets, and motivated adherence. Nine different pharmacists were involved in the MICMI study and adhered to the Danish manual for pharmaceutical care: 'Medication Review—Managing Medicine Manual' [22]. The aim of the

**Fig. 1** Participant flow

‘Medication Review—Managing Medicine’ is to prevent, identify, and resolve DRPs and to contribute to rational pharmacotherapy for patients and society. The pharmacists must have some practical experience or courses in Medication Review’. No further training or standardization was arranged. At 3, 6 and 9 months the same pharmacists telephoned the patients to inquire about the patients’ condition and changes in the medicine, uncover problems and answer questions. Pharmacists could consult the MICMI project physician if they considered a patient’s medication problems to be life threatening. If the physician agreed, the pharmacists should be asked to contact the general

practitioner by fax. There were no standardized criteria for severity of medication problems.

## Outcomes

### Primary outcomes

The primary endpoint was treatment adherence assessed by a pill-count (PC) in all subjects during 1 year. Only oral prescription drugs taken throughout the study period were included in the adherence calculation. At the time of patient inclusion the latter amounted to 1,729 oral drugs

totally in both groups, amongst a total of 3,616 drugs consumed. Cardiovascular drugs represented 69 % of drugs in the adherence calculation. In addition, a project nurse visited all patients initially, then at 6 and 12 months to photograph pills to be counted later by a ‘counter pen’ (a combination of a marker and a digital counter). The adherence rate (%) per drug was calculated as mean adherence rate during 1 year:

$$\frac{(\text{No. pills at first PC} + \text{Subsequent No. pills dispensed} - \text{No. pills at last PC})}{\text{Expected pill consumption}} \times 100$$

We also calculated adherence rates for the intervals of 0–6 and 6–12 months. Adherence rates >100 % were reset to 100 and regarded as full adherence rather than over-medication, as recommended for studies of anti-hypertensive drugs [23]. The mean adherence rate was calculated for each drug consumed with rates <80 % considered as non-adherence.

#### Secondary outcomes

Secondary outcome measures included DRPs, hospitalisations and mortality. DRPs in relation to the patient’s entire medicinal regimen were assessed by pharmacists at home-visits and classified into eight categories [22].

Data on unplanned hospital admissions to medical departments were obtained from the Danish e-Health Portal which holds information on all Danish public hospital admissions. The data search extended to 2 years beyond the end of the study. Mortality data were obtained from the hospital e-journals during the same period. The e-journal is the electronic hospital record. It automatically records information on all deceased patients.

#### Sample size

Sample size calculation for the MICMI study was based on an assumed 20 % medication non-adherence rate and an absolute risk reduction of 10 %. The estimated sample size was 286 patients for each group in the present two-sample comparisons. Power was 90 % with significance set at 0.05. With an assumed dropout risk of approximately 10 % during the follow-up period, a total sample size of 315 patients per patient subgroup was considered meaningful.

#### Randomisation

A total of 945 envelopes (315 per patient subgroup) was prepared with each containing a study inclusion code. At the first home visit by a project nurse, patients were asked to select one envelope.

#### Treatment group ‘blinding’

It was impossible to conceal the identity of patients in the pharmaceutical care group since the procedures were complex and involved the pharmacists and nurses [24].

#### Statistical methods

Statistical analyses were performed using Stata/SE version 12.1. Patient characteristics and differences between patient subgroups were analysed non-parametrically (rank sum test). Categorical data were analysed by contingency tables and mortality data by Cox regression.

#### Approvals and permissions

The Danish Data Protection Agency approved the study.

### Results

#### Patient flow

The temporal flows of pharmaceutical-care and control patients across the study period are shown in Fig. 1. Overall dropout rates were similar to those of patients included in the final analysis with regard to sex, but different in relation to age, i.e. dropouts were older (median age 77; IQR 71–82) than included patients (median age 74; IQR 70–80; rank sum test  $p = 0.0022$ ). More patients’ in the pharmaceutical care group dropped out due to patients’ ‘lack of interest’. The number of other reasons for dropping out was similar in the two groups.

**Table 1** Baseline comparison of pharmaceutical care and control patients

| Variables                       | Pharmaceutical care (n = 253) | Controls (n = 264) |
|---------------------------------|-------------------------------|--------------------|
| Age                             |                               |                    |
| Median                          | 74                            | 74                 |
| IQR                             | 70–80                         | 70–80              |
| Range                           | 65–94                         | 65–91              |
| Sex                             |                               |                    |
| Male                            | 120 (47 %)                    | 130 (49 %)         |
| Female                          | 133 (53 %)                    | 134 (51 %)         |
| Number of prescriptions         |                               |                    |
| Median                          | 7                             | 7                  |
| IQR                             | 5–8                           | 5–8                |
| Range                           | 1–16 <sup>a</sup>             | 3–18 <sup>a</sup>  |
| Type of drugs <sup>b</sup>      |                               |                    |
| Cardiovascular system           | 45 %                          | 44 %               |
| Alimentary tract and metabolism | 13 %                          | 13 %               |
| Blood and blood forming organs  | 13 %                          | 14 %               |
| Nervous system                  | 11 %                          | 11 %               |
| Other                           | 18 %                          | 18 %               |

<sup>a</sup> At the time of inclusion all patients were registered as using five, or more, prescription drugs. At the following interview it appeared that 55 patients were prescribed certain drugs for short-term only, e.g. antibiotics

<sup>b</sup> According to the anatomical therapeutic chemical (ATC) classification system

## Baseline data

Baseline comparisons of pharmaceutical care and control group patients are shown in Table 1.

## Number of drug-related problems (DRPs)

Pharmacists recorded 72 % of DRPs of patients in the pharmaceutical care group (Table 2). The most frequently reported DRPs were adverse drug reactions (22 %), inappropriate drug use by patients (19 %), and subtherapeutic dosages (15 %). General practitioners were contacted by fax on 12 occasions.

## Treatment adherence

The mean treatment non-adherence rate over 1 year did not show any significant difference between the pharmaceutical care and the control groups, i.e. 11 % of pharmaceutical care patients were non-adherent compared to 10 % in the control group (Table 3). A separate analysis of cardiovascular drugs did not show a significant difference between the groups (Table 3).

**Table 2** Number of DRPs identified by a pharmacist during home visits to 253 elderly patients

|                                               | No.        |
|-----------------------------------------------|------------|
| Total drug-related problems                   | 385        |
| Number of patients with drug-related problems | 183 (72 %) |
| Drug-related problems per patient (range)     | 1.5 (0–7)  |
| Untreated indications                         | 48 (13 %)  |
| Improper drug selection                       | 51 (13 %)  |
| Subtherapeutic dosages                        | 58 (15 %)  |
| Inappropriate use by patients                 | 73 (19 %)  |
| Overdosages                                   | 21 (5 %)   |
| Adverse drug reactions                        | 84 (22 %)  |
| Drug interactions                             | 28 (7 %)   |
| Drugs used without indication                 | 22 (6 %)   |
| General practitioner contacts                 | 12         |

Within group comparisons of numbers of non-adherent patients after the first 6 months and after 12 months showed no significant increase in non-adherent patients in either the intervention or the control group (Table 4). In the pharmaceutical care group, 14 % were non-adherent in the first half year compared to 19 % in the last (OR 1.46; 95 % CI 0.91–2.35).

Between groups comparisons showed no significant difference in non-adherence between the pharmaceutical care and the control group after the first or the last 6 months (Table 4).

Non-adherence rates at baseline are unknown as the calculation of non-adherence rates are based on two pill counts with a 6 months' interval and the first pill count was done just after inclusion.

## Hospitalisations

Pharmaceutical care produced no significant effect on total hospital medical admissions at the 24 month follow-up (rank sum test:  $p = 0.47$ ), or the percentages of patients with at least one hospitalisation, i.e. pharmaceutical-care patients 30 % versus control patients 28 % (Table 3). Also, both groups were similar with respect to the number of observation periods (rank sum test:  $p = 0.29$ ), total hospital admissions during the intervention year (rank sum test:  $p = 0.29$ ), and percentages of patients with at least one hospitalisation during the intervention year, i.e. pharmaceutical care group 8 % versus control group 11 % (OR 0.73; 95 % CI 0.41–1.32).

## Mortality

There was no significant impact of pharmaceutical care on mortality at the 24 month follow-up, i.e. pharmaceutical

**Table 3** Estimated risk amongst elderly patients given pharmaceutical care support

|                                   | Pharmaceutical care<br>No. (%) | Controls<br>No. (%) | Odds ratio<br>(95 % CI) | Risk difference<br>(95 % CI) |
|-----------------------------------|--------------------------------|---------------------|-------------------------|------------------------------|
| All drugs <sup>a</sup>            | (n = 253)                      | (n = 264)           |                         |                              |
| Adherent (n = 463)                | 225 (89)                       | 238 (90)            | 1.00                    |                              |
| Non-adherent (n = 54)             | 28 (11)                        | 26 (10)             | 1.14 (0.62, 2.00)       | 1 (−4, 7)                    |
| Cardiovascular drugs <sup>a</sup> | (n = 231)                      | (n = 244)           |                         |                              |
| Adherent (n = 424)                | 210 (91)                       | 214 (88)            | 1.00                    |                              |
| Non-adherent (n = 51)             | 21 (9)                         | 30 (12)             | 0.71 (0.40, 1.29)       | −3 (−9, 2)                   |
| Hospitalisation                   |                                |                     |                         |                              |
| None (n = 367)                    | 176 (70)                       | 191 (72)            | 1.00                    |                              |
| Once or more (n = 150)            | 77 (30)                        | 73 (28)             | 1.14 (0.78, 1.67)       | 3 (−5, 11)                   |
| Mortality                         |                                |                     | Hazard ratio (95 % CI)  |                              |
| Alive (n = 484)                   | 234 (92.5)                     | 250 (95)            | 1.00                    |                              |
| Dead (n = 33)                     | 19 (7.5)                       | 14 (5)              | 1.41 (0.71, 2.82)       |                              |

<sup>a</sup> Only oral prescription drugs taken permanently during the study period were included in the adherence calculation

**Table 4** Estimated risk of non-adherence 0–6 and 6–12 months

|                          | Pharmaceutical care<br>No. (%) | Controls<br>No. (%) | Between groups<br>Odds ratio<br>(95 % CI) |
|--------------------------|--------------------------------|---------------------|-------------------------------------------|
| Non-adherent 0–6 months  | 35 (14)                        | 39 (15)             | 0.93 (0.57, 1.52)                         |
| Non-adherent 6–12 months | 48 (19)                        | 42 (16)             | 1.24 (0.78, 1.95)                         |
| Within groups            |                                |                     |                                           |
| Odds ratio (95 % CI)     | 1.46 (0.91, 2.35)              | 1.09 (0.68, 1.75)   |                                           |

care group deaths 7.5 % versus control group 5.0 % (Table 3).

## Discussion

Pharmacists identified many DRPs, but few necessitated a medical consultation, presumably because pharmacists were asked to contact general practitioners only for severe cases.

Difference of treatment non-adherence between the pharmaceutical care group and control group were not significant. Pill counts could have had a positive effect on adherence in both groups. We were unable to measure baseline treatment adherence. Owing to the randomized controlled trial design, we may consider adherence rates as equal in both groups, but we cannot with certainty conclude that pharmaceutical care made no impact on adherence. The non-adherence rates after the first 6 months versus the last six did not differ significantly in the pharmaceutical care group nor in the control group. The percentage of non-adherent patients calculated for the total study period was not the mean of the two subperiods of 6 months but slightly lower. The same method for

calculating non-adherence rates was used in the 6 and 12 months' periods. The number and type of drugs included in the analyses were also the same. We included only prescribed oral drugs taken throughout the whole study period in the non-adherence rates. However, several factors may influence the results when studying relatively short periods. Patients may have bought new pill boxes that were not shown to the pill-counting nurse just before the 6 months' visit, or a change in dosages ordered by the practitioner may not have been implemented by the patient or not told to the nurse. So, pill counting over longer periods may be preferable as a tool for measuring adherence. Since we compared two groups and found exactly the same tendency, we consider our results as valid. When adherence rates are compared between studies, the length of the observation period may be important.

The adherence rate was high in both groups. This could be due to: (1) PC acting as an intervention with positive effects on treatment adherence if patients, aware of being watched closely, improved their adherence [25]; (2) to a selection bias if solely patients with a positive attitude agreed to participate; or (3), due to the focus of this study being on drugs taken continuously without dosage adjustments.



Before the MICMI study started a 10 % risk reduction was considered meaningful. As the overall adherence rate was found to be 90 % an improvement of 10 % in adherence rate became improbable due to the ceiling effect.

Most similar studies that incorporated a degree of home pharmaceutical care reported positive effects on adherence to long-term therapy [3–7]. Three studies showed no effect, two of which included high baseline adherence rates [8, 9] with the third left unexplained [10]. High baseline adherence allows little room for further improvement. Also, adherence was measured differently [8–10]. Only two of these studies also measured clinical outcomes, such as levels of blood pressure and low-density lipoprotein [3], or the incidence of pulmonary oedema and peripheral oedema [6]. Both studies found improvement of medication adherence and clinical outcomes, but the latter also reported that the objective clinical benefit on oedema may not be noticed by the patients [6]. Follow-up durations, except for one 12 months study [7], were also generally shorter than reported here, i.e. 3 or 6 months [3, 4, 6, 8, 9] and even 3 weeks [5]. Our between groups and within groups analyses at 6 and 12 months did not change our results. One study [9] was similar to ours with regard to its population and interventions. The authors, too, found many DRPs and no effect on adherence. However, its baseline adherence rate was high and in contrast to our study its patients took two or more drugs, and only 114 cases completed the follow-up period [9].

Our results of impact on adherence are in accordance with a Cochrane review concluding that current methods of improving adherence in chronic health problems are mostly complex and not very effective [26]. Pharmaceutical care is a complex intervention, defined as an intervention being made up of ‘various interconnecting parts’ [27]. Since the ‘various interconnecting parts’ of pharmaceutical care are difficult to specify, problems may be encountered in attempting to replicate and evaluate such interventions [27]. But for short-term treatments, quite simple interventions may increase adherence and improve patient outcomes, though less than half of the studies showed any benefit [26].

Pharmacist interventions had no effect on hospitalisation rates or mortality, which agrees with most other reports. Four studies found no difference in hospitalisation rates [8, 14–16] and two found significant or borderline significant [4] hospitalisation effects, one positive [4] and the other negative [17]. Two studies, somewhat similar to ours, reported no effects on hospitalisation, but their hospitalisation rates were small and their follow-up periods lasted only 3–6 months [15, 16], compared to our 24 months. We chose a longer follow-up because our elderly patients were active and many of their drugs were preventive. Four studies found no impact on mortality [4, 8, 14, 17]. Some of the results in the studies reporting effect on

hospitalisation and mortality were too small for meaningful hospitalisation and mortality statistics [4, 14–16]. Moreover, Krska [28] considered hospitalisation insufficiently sensitive for evaluating the impact of pharmacists’ interventions.

The current trend is face-to-face meetings between general practitioners and pharmacists, or multidisciplinary healthcare teams and pharmacists. This could prove to be an important success factor for pharmacist-led interventions [29, 30]. In our case, fax were sent to general practitioner with no further discussion of the DRPs. When pharmacists present their medication review findings and recommendations at face-to-face meetings, the pharmacists are able to facilitate discussions and decision-making with regard to DRPs. On the other hand, the physicians and nurses are able to provide additional clinical information essential for making appropriate interventions [29].

In our case, the few exchanges by fax with prescribers could explain why pharmaceutical care intervention had little impact.

The main strength of our study lies in its large number of the elderly polypharmacy patients living at home and pharmaceutical care provided by nine pharmacists in order to reduce bias.

Our study has certain limitations. First, there is still no ‘gold standard’ for measuring medication adherence [13, 31]. In the MICMI study, pill counts were chosen because of the strength and limitations of this method [13]. We observed that different methods of adherence measuring may give different results [32]. Since we compared two groups using the same methods, we consider our results valid. Second, we did not evaluate medication adherence before inclusion. Third, the high medication adherence level left little scope for further improvement. Fourth, patients hospitalised for more than 7 days during the study were excluded before the final adherence evaluation. These participants may have been more ill, and less adherent and may have benefited more from the pharmaceutical care than those included. Fifth, the sample size calculation relied on medication adherence as its primary outcome variable. The sample size for our primary outcome was not achieved due to a higher number of drop-outs than expected (Fig. 1) and it may also have been underpowered for the other outcomes. If sample size calculations for hospitalisation had been based on the results of Naunton et al. [4], the estimated sample size would have been 179 patients for each group in the present two-sample comparisons with a power of 90 %. But a much higher number of patients would have been needed if the sample size calculations for mortality were based on the results of Wu et al. [18]. The estimated sample size would have been 734 patients for each group in the present two-sample comparisons with a power of 90 %.

Our results may be underpowered, but they do not show any trend toward an effect of pharmaceutical care on adherence, hospitalisation, and mortality.

## Conclusion

Pharmaceutical care can identify DRPs in active elderly patients prescribed polypharmacy, but our control group comparisons failed to show significant differences for treatment adherence, hospitalisation rates, or mortality. Instead of focusing on active elderly patients a more critical target might be ‘patients at-risk’ of serious illness. Also, a more direct communication with general practitioners could be important for successful pharmacist-led interventions. Conversely, evaluating the impact of pharmaceutical care with objective endpoints, such as hospitalisation rates and mortality, may be relevant but would appear to lack sensitivity.

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